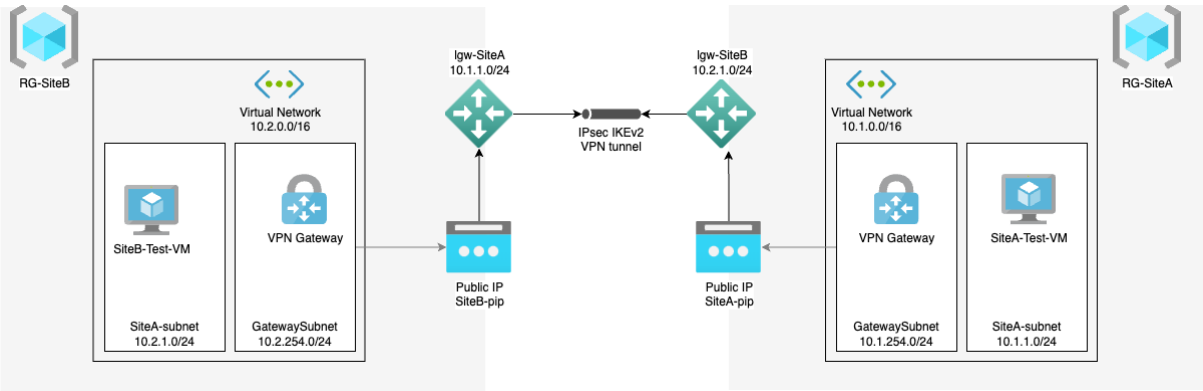
Laboratory work No. 6

Objective – simulate a connection between different cloud instances virtual networks using site to site VPN connection.

Software – portal.azure.com, draw.io

Intro

The aim of this lab work is to simulate networking between two cloud platforms. This time you will have to create two groups of resources in different regions and use IPsec technology to connect the different networks. During this lab you will create different virtual networks (note the addressing in the diagram) and virtual servers between which you will check the connection at the end of the lab.



Tasks

1. Create two different resource groups in two different Azure regions.
2. Within each resource group create virtual network and assign proper IP addresses
3. Create a **gateway subnet** (look for the instructions)
4. Create a **virtual network gateway** (creating a gateway can take 20-45 minutes).
    a. You won‘t need active-active mode
    b. You can use *VpnGw1* SKU
5. Create two VM machines in different resources groups and assign to local virtual network you have created before. Do not create public IP addresses.
6. Create a local network gateway. Be careful. Here you have to setup remote site settings. Remote site public IP address and virtual network subnet.
7. After virtual network gateway and local network gateway setup is done, create a VPN connection:
    a. Connection type – Site-to-site (IPsec)
    b. IKE protocol – IKEv2
8. After both **Connections** are created you have to setup the **same** preshared key. Do it via Setting -> Authentication
9. After the deployment is finished, you can view the connection on the **Connections** page of the virtual network gateway. The status changes from *Unknown* to *Connecting* and then to *Succeeded and Connected*.
10. Check connectivity between virtual machines by using private IP addresses (tip: use serial console feature)
11. Check logs and other metrics
12. Cleanup resources

For more detailed instructions follow: Tutorial site-to-site portal

Questions

1. Explain the purpose of setting up a site-to-site VPN gateway in Azure. How does it benefit organizations?
2. What are the key components required to establish a site-to-site VPN connection in Azure?
3. Why is it necessary to create a virtual network before setting up a VPN gateway?
4. What are the different types of VPN gateways available in Azure, and how do they differ from each other?
5. Explain the role of the shared key in establishing a secure connection between two sites?
6. What troubleshooting steps would you take if the site-to-site VPN connection fails to establish? Provide some common issues and their possible resolutions.
7. How can you monitor the performance and health of a site-to-site VPN connection in Azure? What tools or metrics would you use for monitoring purposes?

In your report, please provide:

- Work progress.
- Answer the questions asked in the course of the work.